

All nonsymmetric convolution-power limits on a compact quantum group

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Abstract

For an arbitrary state on an arbitrary compact quantum group, we classify the weak- $*$ cluster set of its convolution powers. It is a compact monothetic group obtained by simultaneously rotating the peripheral eigenspaces of the finite Peter–Weyl Fourier matrices. This gives an exact criterion for ordinary convergence, identifies the (possibly atypical) idempotent limit, characterizes convergence to Haar, and yields point-norm convergence of the associated convolution operators. No symmetry, traciality, finite dimensionality, or uniform spectral gap is required.

1 The question and the answer

Let $\mathbb{G}$ be a compact quantum group with reduced C^* -algebra $C(\mathbb{G})$, coproduct Δ , and Haar state h . For states ω, ρ write $\omega \star \rho = (\omega \otimes \rho)\Delta$. Franz and Skalski ask in Section 4 of [1] for the limit behaviour of $\phi^{\star n}$, or of the corresponding iterated convolution operators, when no symmetry assumption is made on ϕ . The warning in their question is essential: an idempotent state need not be the Haar state of a quantum subgroup.

Choose, for every $\alpha \in \text{Irr}(\mathbb{G})$, an irreducible unitary corepresentation

$$U^\alpha = (u_{ij}^\alpha) \in B(H_\alpha) \otimes \text{Pol}(\mathbb{G}), \quad \Delta(u_{ij}^\alpha) = \sum_k u_{ik}^\alpha \otimes u_{kj}^\alpha,$$

and define the Fourier contraction

$$A_\alpha = (\text{id} \otimes \phi)(U^\alpha) = (\phi(u_{ij}^\alpha))_{ij} \in B(H_\alpha).$$

Let $\Lambda_\alpha = \sigma(A_\alpha) \cap \mathbb{T}$. For $\lambda \in \Lambda_\alpha$, let $P_{\alpha,\lambda}$ be the spectral projection of A_α at λ . These projections are orthogonal. Put

$$I_\phi = \{(\alpha, \lambda) : \lambda \in \Lambda_\alpha\}, \quad g_\phi = (\lambda)_{(\alpha,\lambda) \in I_\phi} \in \mathbb{T}^{I_\phi},$$

and let

$$K_\phi = \overline{\{g_\phi^n : n \geq 1\}} \subseteq \mathbb{T}^{I_\phi}.$$

The closure is a compact monothetic group (and is the one-point group if I_ϕ is empty).

Theorem 1 (Complete Fourier–peripheral classification). *For each $z = (z_{\alpha,\lambda}) \in K_\phi$ there is a unique state ϕ_z on $C(\mathbb{G})$ satisfying*

$$(\text{id} \otimes \phi_z)(U^\alpha) = B_\alpha(z) := \sum_{\lambda \in \Lambda_\alpha} z_{\alpha,\lambda} P_{\alpha,\lambda} \quad (\alpha \in \text{Irr}(\mathbb{G})). \quad (1)$$

Then:

1. The weak- $*$ cluster set of $(\phi^{\star n})_{n \geq 1}$ is exactly $\{\phi_z : z \in K_\phi\}$.

2. The map $z \mapsto \phi_z$ is a topological group isomorphism from K_ϕ onto that cluster set equipped with convolution: $\phi_z \star \phi_w = \phi_{zw}$. Its identity is ϕ_1 , whose α -Fourier matrix is $\sum_{\lambda \in \Lambda_\alpha} P_{\alpha, \lambda}$.
3. The Cesàro means $N^{-1} \sum_{n=1}^N \phi^{*n}$ converge weak-* to an idempotent state ϕ_C characterized by $(\text{id} \otimes \phi_C)(U^\alpha) = P_{\alpha, 1}$. It is absorbing for the cluster group: $\phi_z \star \phi_C = \phi_C = \phi_C \star \phi_z$ for every $z \in K_\phi$.
4. The ordinary powers converge weak-* if and only if

$$\sigma(A_\alpha) \cap \mathbb{T} \subseteq \{1\} \quad \text{for every } \alpha \in \text{Irr}(\mathbb{G}). \quad (2)$$

In that event $\phi^{*n} \rightarrow \phi_1 = \phi_C$ and $(\text{id} \otimes \phi_1)(U^\alpha) = P_{\alpha, 1}$.

5. One has $\phi^{*n} \rightarrow h$ if and only if

$$\sigma(A_\alpha) \cap \mathbb{T} = \emptyset \quad \text{for every nontrivial } \alpha \in \text{Irr}(\mathbb{G}). \quad (3)$$

For every convergence or cluster net above, the corresponding left or right convolution operators converge point-norm on $C(\mathbb{G})$. In the reduced Haar GNS realization they converge strongly on $L^2(\mathbb{G}, h)$.

Thus the all-compact answer is exact even when the limit is an atypical idempotent. Condition (2) is the aperiodicity condition; condition (3) adds ergodicity.

As a sanity check, take an ordinary compact group and $\phi = \delta_g$. Then the cluster group consists of the point masses on $\overline{\langle g \rangle}$, its identity is δ_e , and ϕ_C is Haar measure on $\overline{\langle g \rangle}$. This also illustrates why the cluster-group identity and the Cesàro idempotent must not be conflated.

2 Finite contraction lemma

Lemma 1. *If A is a contraction on a finite-dimensional Hilbert space, its unit-circle eigenvalues are semisimple, their eigenspaces are mutually orthogonal and reducing, and*

$$A^n - \sum_{\lambda \in \sigma(A) \cap \mathbb{T}} \lambda^n P_\lambda \longrightarrow 0 \quad (4)$$

in norm. Moreover $N^{-1} \sum_{n=1}^N A^n \rightarrow P_1$.

Proof. If $Ax = \lambda x$ and $|\lambda| = 1$, then $\langle (I - A^*A)x, x \rangle = 0$. Positivity gives $A^*Ax = x$, hence $A^*x = \bar{\lambda}x$. The eigenspace therefore reduces A ; eigenspaces for distinct unit-circle eigenvalues are orthogonal. A nontrivial Jordan block at a unit-circle eigenvalue would make (A^n) unbounded, contrary to $\|A^n\| \leq 1$. On the orthogonal complement of all these eigenspaces, the spectrum lies strictly inside the unit disk, so the remaining powers tend to zero. This proves (4). Cesàro averaging retains only the $\lambda = 1$ summand. $\square$

3 Proof of the classification

Slicing a unitary by a state is contractive, so $\|A_\alpha\| \leq 1$. The coproduct formula gives

$$(\text{id} \otimes (\omega \star \rho))(U^\alpha) = (\text{id} \otimes \omega)(U^\alpha)(\text{id} \otimes \rho)(U^\alpha). \quad (5)$$

In particular the Fourier matrix of ϕ^{*n} is A_α^n . Lemma 1 now gives, separately on every Peter–Weyl block,

$$A_\alpha^n - \sum_{\lambda \in \Lambda_\alpha} \lambda^n P_{\alpha, \lambda} \longrightarrow 0. \quad (6)$$

The only point needing care is simultaneous passage over all irreducibles. The closure of the positive powers of one element of a compact group is a compact subgroup: its closure contains the identity (apply compactness to pairs of sufficiently close powers), and hence contains inverses. Thus K_ϕ is a compact group. Moreover every tail of (g_ϕ^n) is dense in K_ϕ , because its closure is $g_\phi^K K_\phi = K_\phi$. Thus every point of K_ϕ is realized by a subnet with $n \rightarrow \infty$.

Fix $z \in K_\phi$ and choose a net (n_i) with $g_\phi^{n_i} \rightarrow z$. By (6), the Fourier matrices of ϕ^{*n_i} converge to $B_\alpha(z)$ for every α . The state space is weak-* compact, so a subnet has a weak-* limit. Every such limit has the Fourier matrices in (1); since the coefficients u_{ij}^α span the norm-dense Hopf *-algebra $\text{Pol}(\mathbb{G})$, these matrices determine the functional uniquely. Hence the original net converges to a unique state ϕ_z . The same density argument shows that $z \mapsto \phi_z$ is continuous and injective.

Conversely, from any weak-* cluster net of (ϕ^{*n}) , compactness of K_ϕ supplies a further subnet on which g_ϕ^n converges to some z ; the preceding paragraph identifies the cluster state with ϕ_z . This proves part 1. Orthogonality of the peripheral spectral projections and (5) give

$$B_\alpha(z)B_\alpha(w) = B_\alpha(zw),$$

so Fourier uniqueness yields $\phi_z \star \phi_w = \phi_{zw}$. Compactness then gives the topological group assertion in part 2.

By Lemma 1, the Fourier matrices of the Cesàro means converge to $P_{\alpha,1}$. Norm density and uniform boundedness of the states give weak-* convergence to a state ϕ_C with those Fourier matrices. Since $P_{\alpha,1}^2 = P_{\alpha,1}$, Fourier uniqueness proves idempotence. The coordinate belonging to the phase $\lambda = 1$ is identically one throughout K_ϕ , so $B_\alpha(z)P_{\alpha,1} = P_{\alpha,1} = P_{\alpha,1}B_\alpha(z)$; this proves the two absorption identities. Notice that the cluster-group identity ϕ_1 retains *all* peripheral eigenspaces, whereas Cesàro averaging retains only the phase-one eigenspace. They coincide precisely when condition (2) holds.

The scalar sequence λ^n , $\lambda \in \mathbb{T}$, converges only when $\lambda = 1$. Equations (6) and density therefore prove part 4. Finally, the Fourier matrix of the Haar state is 1 on the trivial corepresentation and 0 on every nontrivial irreducible. This proves part 5.

For the operator claim, on each coefficient space convolution is multiplication by the relevant Fourier matrix. Hence the asserted operator convergence is norm convergence on $\text{Pol}(\mathbb{G})$. Convolution by a state is contractive; approximating an arbitrary element by coefficient polynomials proves point-norm convergence on $C(\mathbb{G})$. The Schwarz inequality and Haar preservation give contractions on the Haar GNS L^2 space, and density of the coefficient algebra proves strong convergence there.

4 A support-free periodicity test

The peripheral obstruction admits an intrinsic equality-case description. Let $(\pi_\phi, H_\phi, \Omega_\phi)$ be the GNS representation of ϕ and put

$$V_\alpha = (\text{id} \otimes \pi_\phi)(U^\alpha), \quad J_\alpha \xi = \xi \otimes \Omega_\phi.$$

Then $A_\alpha = J_\alpha^* V_\alpha J_\alpha$.

Corollary 1 (Deterministic-vector criterion). *For $\lambda \in \mathbb{T}$ and $0 \neq \xi \in H_\alpha$,*

$$A_\alpha \xi = \lambda \xi \iff V_\alpha(\xi \otimes \Omega_\phi) = \lambda \xi \otimes \Omega_\phi.$$

*Consequently ϕ^{*n} converges precisely when every deterministic corepresentation vector has phase 1, and it converges to Haar precisely when no nontrivial irreducible corepresentation has any deterministic vector.*

Proof. The reverse implication follows by compression. For the forward implication,

$$\|J_\alpha^* V_\alpha J_\alpha \xi\| = \|\xi\| = \|V_\alpha J_\alpha \xi\|.$$

Equality for the orthogonal projection $J_\alpha J_\alpha^*$ forces $V_\alpha J_\alpha \xi$ to lie in the range of J_α . Its compression is $\lambda \xi$, so it equals $\lambda J_\alpha \xi$. $\square$

For an ordinary compact group, this says that a unitary representation vector is multiplied by one fixed phase almost everywhere on the support of the driving measure—the familiar periodic-coset obstruction. In the quantum case it remains meaningful without forcing the limiting idempotent to arise from a subgroup.

Remark 1. McCarthy’s finite-dimensional theorem [2] translates irreducibility and periodicity into quasi-subgroups and cyclic cosets. Theorem 1 instead works for every compact quantum group and records all cluster states, including infinitely many phases and non-Haar idempotent limits. Related fixed-point and Cesàro theory for locally compact quantum groups appears in [3].

References

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